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Inventor(s)

GROSS; Susan Joy

SYSTEMS AND METHODS FOR DATA ACCESS MANAGEMENT

Abstract

Aspects of the technical solutions described herein relates to a system for data access management. A system receives, responsive to an interaction with an information resource of a provider portal, a request to access data. The system authenticates the user. The system provides, to a data source, a request to grant, to the system, access to user data of the user stored at one or more data sources. The system receives an access token to allow the system to retrieve user data from the data sources. The system transmits a request to access the user data using the access token assigned to the system. The system receives the user data responsive to transmitting the request to access the user data. The system transmits, responsive to a request from the partner, a portion of the user data based on values corresponding to a type of data obtained at registration of the provider portal.

Inventors: GROSS; Susan Joy (Bronx, NY)

Applicant: GROSS; Susan Joy (Bronx, NY)

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS [0001] The present application claims the benefit of and priority to U.S. Provisional Application No. 63/560,912 filed on Mar. 4, 2024, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

[0002] The present disclosure relates generally to the field of data access and data management, specifically data access between different data sources including information about users across each data source.

SUMMARY

[0003] Aspects of the present disclosure relates to a system that includes one or more processors of a data management system coupled to memory. The one or more processors of the data management system may receive, responsive to an interaction with an actionable object presented within an information resource of a provider portal at a client device of a user, a request to access data. In some embodiments, the one or more processors may authenticate, responsive to receiving authentication credentials from the client device, the user of the client device. In some embodiments, the one or more processors may provide, to a data source, a request to grant, to the data management system, access to user data of the user stored at one or more data sources, the request including a client identifier identifying the data management system. In some embodiments, the one or more processors may receive, responsive to the request to grant, an access token to allow the data management system to retrieve user data of the user from the one or more data sources. In some embodiments, the one or more processors may transmit, via one or more application programming interfaces (API) of the one or more data sources, a request to access the user data of the user using the access token assigned to the data management system. In some embodiments, the one or more processors may receive the user data via the one or more APIs responsive to transmitting the request to access the user data. In some embodiments, the one or more processors may store the user data in one or more data structures managed by the data management system. In some embodiments, the one or more processors may transmit, responsive to a request from one or more servers of the provider portal, a portion of the user data of the user to the one or more servers of the provider portal based on one or more values corresponding to a type of data obtained at registration of the provider portal with the data management system.

[0004] In some embodiments, the one or more processors may receive, responsive to a second interaction with a second actionable object presented within a second information resource of a second provider portal at the client device of the user, a request to access data. In some embodiments, the one or more processors may authenticate, responsive to receiving authentication credentials from the client device, the user of the client device. In some embodiments, the one or more processors may provide, to a second data source, a second request to grant, to the data management system, access to user data of the user stored at the second data source, the request including the client identifier identifying the data management system. In some embodiments, the one or more processors may receive, responsive to the second request to grant, a second access token to allow the data management system to retrieve user data of the user from the one or more data sources. In some embodiments, the one or more processors may transmit, via one or more application programming interfaces (API) of the one or more data sources, a second request to access the user data of the user using the second access token assigned to the data management system. In some embodiments, the one or more processors may receive the user data via the one or

more APIs responsive to transmitting the second request to access the user data. In some embodiments, the one or more processors may store the user data in one or more data structures managed by the data management system. In some embodiments, the one or more processors may transmit, responsive to a second request from one or more servers of the second provider portal, a second portion of the user data of the user to the one or more servers of the second provider portal based on one or more values corresponding to a type of data obtained at registration of the second provider portal with the data management system.

[0005] Aspects of the present disclosure relates to a computer implemented method. The method may include receiving, by one or more processors, responsive to an interaction with an actionable object presented within an information resource of a provider portal at a client device of a user, a request to access data. In some embodiments, the method may include authenticating, by the one or more processors, responsive to receiving authentication credentials from the client device, the user of the client device. In some embodiments, the method may include providing, by the one or more processors, to a data source, a request to grant, to the data management system, access to user data of the user stored at one or more data sources, the request including a client identifier identifying the data management system. In some embodiments, the method may include receiving, by the one or more processors, responsive to the request to grant, an access token to allow the data management system to retrieve user data of the user from the one or more data sources.

[0006] In some embodiments, the method may include transmitting, by the one or more processors, via one or more application programming interfaces (API) of the one or more data sources, a request to access the user data of the user using the access token assigned to the data management system. In some embodiments, the method may include receiving, by the one or more processors, the user data via the one or more APIs responsive to transmitting the request to access the user data. In some embodiments, the method may include storing, by the one or more processors, the user data in one or more data structures managed by the data management system. In some embodiments, the method may include transmitting, by the one or more processors, responsive to a request from one or more servers of the provider portal, a portion of the user data of the user to the one or more servers of the provider portal based on one or more values corresponding to a type of data obtained at registration of the provider portal with the data management system.

[0007] In some embodiments, the method may include receiving, by one or more processors, responsive to a second interaction with a second actionable object presented within a second information resource of a second provider portal at the client device of the user, a second request to access data. In some embodiments, the method may include authenticating, by the one or more processors, responsive to receiving authentication credentials from the client device, the user of the client device. In some embodiments, the method may include providing, by the one or more processors, to a second data source, a second request to grant, to the data management system, access to user data of the user stored at a second data source, the request including the client identifier identifying the data management system. In some embodiments, the method may include receiving, by the one or more processors, responsive to the second request to grant, a second access token to allow the data management system to retrieve user data of the user from the one or more data sources.

[0008] In some embodiments, the method may include transmitting, by the one or more processors, via one or more application programming interfaces (API) of the one or more data sources, a second request to access the user data of the user using the second access token assigned to the data management system. In some embodiments, the method may include receiving, by the one or more processors, the user data via the one or more APIs responsive to transmitting the second request to access the user data. In some embodiments, the method may include storing, by the one or more processors, the user data in one or more data structures managed by the data management system. In some embodiments, the method may include transmitting, by the one or more processors, responsive to a second request from one or more servers of the second provider portal, a second

portion of the user data of the user to the one or more servers of the second provider portal based on one or more values corresponding to a type of data obtained at registration of the second provider portal with the data management system.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- [0009] FIG. 1 depicts a block diagram of a system for secure access to data, in accordance with an embodiment;
- [0010] FIG. 2 depicts a block diagram of an interaction system for secure access to the data to a partner server, in accordance with an embodiment;
- [0011] FIGS. 3A-3C depict a user interface of a user device running on an application, in accordance with an embodiment;
- [0012] FIGS. 4A-4C depict another user interface on the application of the client device, in accordance with an embodiment;
- [0013] FIG. 5 depicts a data structure of an electronic health record, in accordance with an embodiment;
- [0014] FIG. 6A depicts an example user interface for managing access for the partners, in accordance with an embodiment;
- [0015] FIG. 6B depicts an example user interface for revoking consent to the partners, in accordance with an embodiment;
- [0016] FIG. 7 depicts a system for an exchange of requests between the system and the partner server, in accordance with an embodiment;
- [0017] FIG. 8 depicts a flowchart to implement secure access to the data via SDK configuration, in accordance with an embodiment;
- [0018] FIG. 9 depicts an example computing system, in accordance with an embodiment.

DETAILED DESCRIPTION

[0019] Today, many users and patients need to use multiple software applications to view their healthcare data. This can be cumbersome and inefficient, as it requires navigating through different interfaces and systems. Additionally, users face challenges in selectively sharing their healthcare information with specific organizations. This selective sharing allows for the user to retain control and for maintaining privacy and ensuring that only relevant information is shared with authorized entities. Existing technological solutions that allow for data sharing across organizations fail to provide a robust system through which data stored in one or more databases can be selectively shared based on the organization and/or the user's permissions. As a result, existing technological solutions are at risk of data leakage and fail to provide adequate privacy controls when transmitting data to other systems.

[0020] The technical solutions described herein overcomes these technical challenges by establishing limits on the types of data that can be shared with provider portals based on limits at registration and relying on access token management to provide fine-grained access control to users. The system includes an authorization server, allowing users to provide fine-grain access control of their healthcare information to organizations that request it. This ensures that users can selectively share their data with authorized entities while maintaining privacy. In addition, the system also uses an access token mechanism to authenticate users and grant access to their data. The authentication code is passed back to the server in exchange for an access token, which allows the user to retrieve and share their data with partners, prior to the expiration of the access token. This access token mechanism ensures secure and controlled access to user data, addressing the challenges of data sharing and privacy. Overall, the technical solutions described herein can provide a robust and efficient solution for managing and sharing healthcare information, reducing

data leakage, and improving privacy of sensitive information that conventional systems often struggle to provide.

[0021] The technical solutions described herein addresses technical challenges by setting predefined limits on the types of data that can be shared with provider portals during registration and utilizing access token management for fine-grained user access control. The system features an authorization server that enables users to grant precise access control over their healthcare information to requesting organizations, ensuring selective data sharing while maintaining privacy. Additionally, the system employs an access token mechanism to authenticate users and grant data access. The authentication code is exchanged for an access token, allowing users to retrieve and share their data securely with partners. This access token mechanism ensures secure and controlled data access, effectively addressing data sharing and privacy challenges. The technical solution described herein can efficiently manage and share healthcare information, overcoming the technical limitations of previous systems.

[0022] Referring now to FIG. 1, FIG. 1 depicts a block diagram of a system **100** for secure data access management. In an overview, the system **100** may include at least one data management system **102**, a set of user devices **104A-N** (hereinafter generally referred to as “user devices **104**” or as a “user device **104**”), data sources **112A-N** (hereinafter generally referred to as “data sources **112**” or as a “data source **112**”), and partner servers **114A-N** (hereinafter generally referred to as “partner servers **114**” or as a “partner server **114**”), communicably coupled to at least one partner database **116**. The various components of the system **100** are communicably coupled over at least one network **101**. At least one user device **104** may include at least one application **106**. The application **106** may provide at least one information resource **108** using one or more software development kits **110A-N** (hereinafter referred to as “SDKs **110**” or as an “SDK **110**”).

[0023] The application **106** can be a browser. In some embodiments, the application **106** can be a mobile application. In some embodiments, the application be an application provided by a partner of the data management system.

[0024] The one or more networks **101** include any hardware and software capable of conducting communication between the data management system **102** and one or more user devices **104**, and one or more databases **118**. The network **101** may include hardware and software, internal or external to a computing infrastructure, which may include any number of public networks and/or private computing networks through which the user devices **104** communicates with the data management system **102**, the data sources **112**, and the partner server **114**. The components of the network **101** may host or conduct inter-device communications in accordance with various communication protocols, but not limited to, Transmission Control Protocol and Internet Protocol (TCP/IP), User Datagram Protocol (UDP), and IEEE communication protocols. Non-limiting examples of the network(s) **105** include, but are not limited to, Local Area Network (LAN), Wireless Local Area Network (WLAN), Metropolitan Area Network (MAN), Wide Area Network (WAN), and the Internet.

[0025] The data management system **102** may include at least one API manager **120**, Fast Healthcare Interoperability Resources (FHIR) manager **122**, interaction manager **124**, data source manager **126**, authorization manager **128**, and a profile generator **130**. The data management system **102** may have access to at least one database **118**.

[0026] The database **118** may store, maintain, or otherwise include one or more user profiles **132A-N** (hereinafter generally referred to as profiles **132**), one or more tokens **134**, and FHIR data **136**, among others. The functionality of an application **106** may be performed in part on the data management system **102**.

[0027] In further detail, the data management system **102** may (sometimes herein generally referred to as a computing system or a server) be any computing device comprising one or more processors coupled with memory and software and capable of performing the various processes and tasks described herein. The data management system **102** may be in communication with the one or more

user devices **104** and the database **118** via the network **101**. The data management system **102** may be situated, located, or otherwise associated with at least one server group. The server group may correspond to a data center, a branch office, or a site at which one or more servers corresponding to the data management system **102** is situated.

[0028] Referring now to FIG. 1 and FIG. 2, FIG. 2 depicts a block diagram of an interaction system **200** for secure access to the data to a partner server **114**. A user **202** may initiate an interaction **204** with the user device **104**. The user device **104** can execute, monitor, or run the application **106** and provide at least one information resource **208** executing an SDK **110** that is provided by one or more servers associated with the data management system. The SDK **110** can be a script, code, or other executable code that can be inserted into an information resource to enable a user interacting with the information resource to login and interact with the data management system.

[0029] The interaction **204** may include a keypress, a touch on a user interface, a login, a button click, or any other event associated with the presentation of an actionable object (e.g., button, checkbox, etc.) within the information resource **108** as shown in FIG. 2. In some embodiments, the actionable object can be provided via the SDK. The actionable object, when interacted with, can cause the SDK executing within the information resource to provide one or more user interfaces shown in FIGS. 3A-3C.

[0030] FIGS. 3A-3C depict a user interface **300** of the user device **104** running the application **106**. The user interface **300** includes the actionable object **302** within the information resource **208**. The information resource **208** is provided by a server corresponding to a provider portal. For example, the actionable object **302** may be a button to connect the partner server **114** with the data management system **102**. In response to the user **202** interacting with the actionable object **302**, the actionable object **302** may trigger or cause the SDK **110** to provide the user interface as shown in FIGS. 4A-4C.

[0031] FIG. 3B depicts a user interface **330** within the application **106**. The SDK **110** may generate, create, or trigger the user interface **330** within the application **106**. The user interface **330** may include one or more actionable objects **332** which trigger an SDK **110**. The user interface **330** is presented in response to interacting with the actionable object **302** presented on the user interface **300**.

[0032] The user interface **360** may prompt the user **202** to log in to the application **106** using a username and password. The username may be an identifier for the user **202** such as, an electronic mail address, a form of the user's first and last name, or unique ID generated by the data management system **102**. If the user **202** does not have a profile **132** within the database, the user interface **360** may prompt a new user **202** to interact with a second actionable object **364**. The second actionable object **364** may prompt the new user **202** to generate or create the username and password and insert information associated with the new user **202**. For example, when the new user **202** interacts with the second actionable object **364**, the user interface **360** may prompt the new user **202** to generate the username (i.e., test1@gmail.com) and the password (i.e., test123456). During a registration process of a new user of the data management system **102**, the user **202** may input their first name, last name, date of birth, medical insurance, social security number, electronic mail address, medical history, current medications, hospitals, providers, among others.

[0033] A profile generator **130** of the data management system **102** may generate, create, or determine a profile **132** suitable for the new user **202**. The profile **132** may include the username, password, and information associated with the new user **202**. For example, the profile generator **130** may use the username, password, date of birth, and insurance of the new user **202** to generate the profile **132** for the new user **202**. The profile generator **130** may store the profile **132** in the database **118**. The profile **132** may include an association to tokens **134** and FHIR data **136**. The tokens **134** correspond to a number of access tokens **134** which correspond to the one or more data sources **112**. For example, the profile **132** may include an access token **134** to access a data source **112A** during a first time period and a second time period. The access token **134** may allow the user

202 to access the respective data source **112A**. To improve security, the token **134** may be restricted to the respective data source **112** to prevent the user **202** from accessing the wrong data source **112**. [0034] Referring back to FIG. 3C, once the user **202** interacts with a first actionable object **362** within the application **106**, the user device **104** may transmit or send the login credentials to the authorization manager **128** via the user interface **360** that is provided by the SDK **110**. The authorization manager **128** can verify the login credentials of the user **202** by fetching or retrieving the profile **132** associated with the user **202** from the database **118**. For example, after the user **202** interacts with the second actionable object **364**, the authorization manager **128** may compare the username and password of the user interface **360** with the username and password of the profile **132** of the user **202**. Upon a successful match, the authorization manager **128** can trigger or cause the user device to present user interface **400** as shown in FIG. 4A. The authorization manager **128** can be a separate authorization server with which the data management system communicates. In some embodiments, the authorization server can support the OAuth2.1 standard.

[0035] FIG. 4A depicts the user interface **400** within the application **106** with one or more actionable objects **402**. The user interface **400** can indicate, identify, or display one or more aspects of the FHIR data **136** (e.g., medical record, family medical history, demographics, vital signs, among others) that the provider portal is seeking to access from the user's FHIR data. The one or more aspects of the FHIR data **136**, also known as scopes, define the type of FHIR data that the provider portal is seeking access to. The type of FHIR data can include demographic information, medical conditions, medications, medical records, medical history, immunization records, practitioners (e.g., physicians, doctors, nurses), clinical notes, among other aspects of FHIR data. [0036] The scopes of the FHIR data that the provider portal is seeking to access from users' FHIR data is based on a partner registration process. The partner registration process entails the partner registering with the data management system in accordance with the user. During the registration process, the partner submits one or more scopes of the FHIR data that the partner would be allowed to access. For example, the partner can submit that the one or more scopes include at least immunization records and family medical history. In another example, the partner can submit that the one or more scopes include medical allergies, demographics, vital signs, and past hospital visits. In this manner, the partner can have fine-grained access to portions of FHIR data associated with the user. Upon registration, the data management system **102** can store, in a data structure, permissions granted to the partner based on the scopes, as will be discussed in FIG. 5.

[0037] Referring back to FIG. 4A, the user **202** may submit a request to enable the partner to access medical history, medical records, allergies, family history, demographics, and vital signs and other types of FHIR data **136**. In response to a user selecting the actionable object **402** to allow the partner to connect with the electronic health records of the user, the user device presents a second user interface **430** that provides one or more hospitals or other electronic health record data sources for selection. Responsive to a user interacting with the user interface **430**, the user provides consent to enable the data management system to obtain access to the selected electronic health records.

[0038] Within the FHIR data **136** to transmit the FHIR data **136** to the partner server **114**. The user **202** may execute the interaction **204** with the one or more actionable objects **402** to access the FHIR data **136** associated with the one or more data sources **112**. For example, the user **202** can interact with the actionable object **402** to browse a list of hospitals, clinics, service providers, or physicians associated with one or more data sources **112** as shown in FIG. 4B.

[0039] FIG. 4B depicts a user interface **430** with a list of data sources **112** within the application **106**. In response to the user **202** interacting with the actionable object **402**, a SDK **110** can generate, create, or display the user interface **405** and one or more actionable objects **432**. Each actionable object **432** in the actionable objects **432** can interact, identify, access, or connect with the respective data source **112**. For example, a first actionable object **432** can connect the partner sever **114** with the data source **112B**. In another example, a second actionable object **432** can connect the partner sever **114** with the data source **112C**. In yet another example, a third actionable object **432**

can connect the partner sever **114** with the data source **112D**. Upon the selection of one of the actionable objects **432**, the data source manager **126** can connect or access the respective data source **112** to retrieve the FHIR data **136** corresponding to the actionable object **402**. The data source manager **126** can transmit a signal to the SDK **110** to generate a success indication as shown in the user interface **460** in FIG. 4C.

[0040] Referring again to FIG. 1 and FIG. 2, the data management system **102** may receive a request to access data. The request to access FHIR data may be received from a user device. A user of the user device may access a provider portal via an application executing on the user device and select an actionable object on an information resource of the provider portal. In some embodiments, the provider portal can be a website or a mobile application of the partner and the information resource can be a page on the website or the mobile application. The user device **104** may generate the request to access the data based on a user performing an action on the corresponding information resource **108**. The information resource **108** may include an actionable object that is presented on the information resource via an SDK or other script. The SDK or script can be provided by or made available by or on behalf of the data management system and may be used to be included in the website or the mobile application of the partner. The SDK or script can be configured such that an actionable object is provided on the information resource, such that when a user performs an action (e.g., click, tap), on the actionable object, the request to access data is generated. The data to be accessed can a subsequent page or additional content. For example, the user **202** execute an interaction **204** with the information resource **108** of the application **106**. The SDK **110** may include the actionable object and detects the interaction **204** with the actionable object. The SDK **110** can generate the request to access the data **136** associated with the actionable object and transmit the request to the data management system **102**.

[0041] Responsive to receiving the request to access data via the actionable object, the data management system or one or more servers can transmit additional information resources for presentation on the user device. The one or more additional information resources can include fields via which the user can provide authentication credentials corresponding to the data management system to authenticate the user. The authentication credentials may include at least one of the username and the password, biometric data, security tokens, smart cards, Personal Identification Number (PIN), Digital Signatures, Biographic information, among others. The data management system **102** can receive the authentication credentials from the user device **104**. For example, the user device **104** may transmit the authorization credentials to the data management system **102**. In some embodiments, the user device **104** can provide the authentication credentials within the request to access the data. For example, the SDK **110** of the information resource may store the authorization credentials as metadata within the request. In this manner, the data management system **102** can extract the metadata from the request and extricate the authorization credentials.

[0042] The interaction manager **124** of the data management system can receive the request from the user device **104** and examine the request and determine the information resource **108** associated with the actionable object. For example, the interaction manager **124** can receive the request from the user device **104** and determine that the information resource **108** corresponds to the partner from the request. Furthermore, the interaction manager **124** and application programming interface (API) manager **120** can provide the application **106** with one or more user interface elements to induce an interaction **204** based on the previous interactions **204** of the user **202**. For example, the user **202** may prefer check boxes instead of buttons. In this manner, the API manager **120** and interaction manager **124** can trigger the SDK **110** to provide check-boxes to the user **202** as discussed in FIGS. 3A-3C.

[0043] The interaction manager **124** can receive the request from the user device **104** and examine the request and determine a second information resource associated with a second actionable object. For example, the interaction manager **124** can receive the request from the user device **104**

and determine the second information resource of a second partner server **114** corresponding to the second actionable object. Furthermore, the interaction manager **124** and application programming interface (API) manager **120** can provide the application **106** with one or more user interface elements to induce a second interaction **204** based on the previous interactions **204** of the user **202**. For example, the user **202** may prefer buttons instead of checkboxes. In this manner, the API manager **120** and interaction manager **124** can trigger the SDK **110** to provide buttons to the user **202**. In some embodiments, the interaction manager **124** may retrieve FHIR resource information and information corresponding to the user **202** from the application **106**. The interaction manager **124** may upload the FHIR resource information and information corresponding to the user **202**, to an external FHIR server or the data management system **102**. In some embodiments, the interaction manager **124** may convert the information corresponding to the user **202** into an FHIR resource.

[0044] The authorization manager **128** of the data management system can authenticate, validate, or verify the user **202** of the user device **104**. To authenticate the user **202**, the authorization manager **128** can analyze the authentication credentials of the user **202** by matching the authentication credentials with the corresponding profile **132** of the user **202**. The profile **132** can include information about the respective user **202** such as, FHIR Data **136**, tokens **134**, a username and password, a user ID, one or more healthcare providers, one or more partners (e.g., websites, applications, etc.), basic identification information, and interaction history. The user **202** may enter the authentication credentials (e.g., username and password) to the application **106** to access the information resource **108**. The authorization manager **128** may receive the authentication credentials from the user device **104** and verify that the authentication credentials entered in the user device **104** match the authentication credentials within the profile **132**.

[0045] Upon the authenticating the authentication credentials, the data source manager **126** can provide, transmit, or send a request to grant the data management system **102** access to the FHIR data **136** associated with the user **202** stored at the one or more data sources **112**. The data source manager **126** may receive one or more codes **212** from the authorization manager **128** based on a successful match of authentication credentials from the user device **104** and within the database **118**. The codes can be at least one of a flag, a notification to transmit the request, or an alphanumeric collection of values. The data source manager **126** can include memory to maintain a record of each data source **112** for the respective user **202** or respective profile **132**. For example, the data source manager **126** may transmit a request to one or more servers that store data corresponding to a hospital to grant the data management system **102** access to the FHIR data associated with the data source manager **126**. In some embodiments, the data source manager **126** can trigger the authorization manager **128** to use a token generator **210** to generate a request token to transmit to the data sources **112**. The data source manager **126** can embed the codes **212** within the request token to identify the data management system **102** to the data sources **112**. The benefit of the solution described is that a secure system can access and exchange data between the user **202** and the data sources **112**, by using the data management system **102** as an intermediary.

[0046] The data source manager **126** may receive an access token from the data sources **112**. The access token can be a security token, indicator, or flag for allowing access to specific FHIR data **136** for a provider while the access token is active. The provider can be denied access, responsive to an expiration of the access token. The access token can at least include a header, a payload, and a signature, among other aspects to specify the contents of the access token. Prior to receiving the access token, the data source manager **126** can provide the access token to the data source **112** upon completion of registration by the user. Upon receiving the access token, the data source manager **126** can store the access token in the tokens **134** associated with the profile **132** of the user **202**. By storing the access token in the tokens **134**, the data source manager **126** may extract the access token for future use by the user. For example, at a first time period, the data source manager **126** can transmit the request to the data sources **112** to grant access to the data management system **102**. The data source manager **126** can refresh, ping, or query the access token for a time period to

verify, detect, or otherwise identify the expiration of the access token. The time period can be between 4-24 hours and the data source manager **126** can cause the access token to expire if the time period is greater than a specified time period (e.g., 8 hours, 24 hours). The data sources **112** can transmit the access token to the data source manager **126** to provide access to the FHIR data **136** associated with the user **202**. The access token can allow the data management system **102** to retrieve the FHIR data **136** of the user from the one or more data sources **112**, while the access token is active.

[0047] The data source manager **126** can subsequently transmit a refresh token to the data source **112**. The refresh token can be a token, a flag, an indicator, or a marker which indicates that a permission for the data source **112** to access the FHIR data **136**. The refresh token can specify a respective data source **112** for access to the FHIR data **136**. The refresh token can be active or expired. The user can specify when the refresh token is active and when the refresh token expires. The refresh token can be transmitted to the data source **112** in response to the user specifying the data source **112** during the registration process. While the refresh token is active, the data source **112** and the data source manager **126** can transmit and receive the access tokens to access portions of the FHIR data **136**. Upon expiration of the refresh token, the data source **112** may not have access to any FHIR data **136** of the user. For example, upon registration of the user, the user can specify that a first data source **112** can access the demographics and vital signs for a given time period (e.g., one month, two months, three months, one year, two years, etc.) and the data source manager **126** can provide the refresh token to the first data source **112**. In some instances, the refresh token can expire in response to a condition such as a change in insurance, an indication from the user, change in provider, change in home address, etc. Upon expiration of the time period, the refresh token can subsequently expire, thereby revoking access to the FHIR data **136** from the data source **112** or the partner.

[0048] In this manner, the technical solutions described herein can prevent the need to make multiple requests or system calls (e.g., API calls) to obtain the tokens (e.g., refresh token) thereby reducing latency, reducing utilization, improving computer performance, reducing wasted computing resources, and saving memory by utilizing the access tokens and the refresh tokens. For example, the refresh token can remove a need for the data source **112** to generate and transmit API requests to obtain a new token each time the data source **112** may access the FHIR data **136** of the user. Furthermore, the use of the access tokens in this manner can prevent the unnecessary generation and storage of subsequent access tokens while the refresh token is active. For example, the database **118** and the data sources **112** can maintain an expired access token while the refresh token is active. In this manner, the data management system **102** can re-activate the access token when needed on request by the data source **112** thereby saving computing resources that would be otherwise wasted on the generation of a new access token. The technical solutions described herein can improve security using the refresh tokens and access tokens, thereby preventing data leakages, enhancing system performance, preventing unauthorized access, maintaining data integrity, and reducing data loss. For instance, the expiration of the access token after a given time period (e.g., can prevent a data source from viewing FHIR data (e.g., medical history, insurance, etc.) after, for example, four hours, five hours, six hours, 10 hours, 24 hours etc., without the submission of a request to the data management system **102**. Furthermore, the expiration of the refresh token can block, prevent, or otherwise revoke access to all FHIR data associated with the user from the data source or the partner. The expiration of the refresh token can be specified by the user, thereby giving more control to the user regarding the use and distribution of the medical data.

[0049] In some embodiments, the system can include one or more data management systems **102**. Each data management system **102** may search or traverse endpoints associated with the one or more data management systems **102** and save the endpoints as a file (e.g., document, PDF, data structure, storage object). In some embodiments, the data management system **102** may save the endpoints to retrieve the FHIR data **136** from the database **118**. In some embodiments, the data

management system **102** may update information about the user **202** in response to a trigger from the authorization manager **128**. In some embodiments, the interaction manager **128** may trigger a user device **104C** to display an application **106** to provide the user **202** with data associated with the user **202**. In some embodiments, one data management system **102** may transmit a request to each endpoint to retrieve FHIR resources in a centralized data management system **102**. In some embodiments, the data management system **102** may store non-FHIR resource information or sources in the database **118**.

[0050] The data source manager **126** and the API manager **120** may transmit a request to access the FHIR data **136** within the data sources **112** corresponding to the user **202**. To retrieve the FHIR data **136**, the data source manager and the API manager **120** may transmit the access token along with the request. The API manager **120** may use one or more API's corresponding to the data sources **112** to transmit the access token. For example, an API can provide a standardized way to format the FHIR data **136** of the data source **112** to match the format of the FHIR data **136** within the profile **132** of the user **202**. In another example, an API can access the FHIR data **136** from the data sources **112**. The access token can be unique for each profile **132** in the database **118** to prevent the wrong user **202** from accessing the wrong FHIR data **136**. For example, if the data source manager **120** transmits an access token that does not correspond to the respective user **202**, the API manager **120** can use one or more APIs to the data sources **112** to block the request to access the FHIR data **136** within the data source **112**. In response to the data sources **112** receiving the request, the data sources **112** can transmit the FHIR data **136** to the data source manager **126** and the API manager **120**. In some embodiments, the data source manager **126** and the API manager **120** may transmit a second request to access the FHIR data **136** within the data sources **112** corresponding to the user **202**.

[0051] The data source manager **126** and the API manager **120** may receive, access, or retrieve the FHIR data via the one or more APIs of the data sources **112**. Upon reception of the FHIR data, the data source manager **126** and the API manager **120** may store the FHIR data **136** in one or more data structures managed by the FHIR manager **122**. The FHIR manager **122** may generate one or more data structures to hold the FHIR data **136**. The FHIR manager **122** can store the one or more data structures within the profiles **132**. Referring now to FIG. 5, FIG. 5 depicts an example data structure **500** generated by the FHIR manager **122** and stored within the profile **132**. The data structure **500** can include one or more user ID's **502A-N** (generally referred to as a "user ID **502**" or as "user IDs **502**"), one or more healthcare sources **504A-N** (generally referred to as a "healthcare **504**" or as "healthcare sources **504**"), **506A-N** (generally referred to as a "scope **506**" or as "scopes **506**"), and partners **508A-N** (generally referred to as a "partner **508**" or as "partners **508**"). The partners **508** may be described in a similar manner to the partner server **114** described above. In some arrangements, the partners **508** may be the names of an organization hosting the partner server **114**.

[0052] The profile generator **130** can generate, create, or determine the user ID **502** by executing one or more functions to create randomize a collection of alphanumeric values. The user ID **502** may include the collection of alphanumeric values to distinguish and identify each user **202** within the profiles **132**. For example, a first user **202** may have "JS1293" as a user ID **502A**. In another example, a second user **202** may have "JD1680" as a user ID **502B**. Each user ID **502** can correspond to the one or more healthcare sources **504** indicating that one or more users **202** may attend similar healthcare sources **504**. For example, a first user **202** with a user ID **502A** and a second user **202** with user ID **502B** may attend the same healthcare source **504** (e.g., hospital, clinic, physician), therefore the healthcare source **504** may include FHIR data **136** for the first user **202** and the second user **202**. The healthcare sources **504** may be described in a similar manner as the data sources **112** because data management system **102** interacts with the online server (e.g., Cloud server, Web server, Database server, proxy server, etc.) of the healthcare sources **504**. For example, the data source manager **126** may retrieve and transmit the tokens **134** to an online server

of the healthcare sources **504** or otherwise the data source **112** corresponding to the healthcare sources **504**.

[0053] The FHIR data **136** can include a list, table, matrix, or collection of the scopes **506**. The scopes **506** can indicate one or more portions of the FHIR data **136** that a partner **508** may access. For example, the FHIR data **136** can include allergy records, medication history, vital signs, demographics, and family conditions. The scopes **506A-C** can include family conditions, medication history and demographics. Therefore, partners **508** can access the scopes **506A-C** of the FHIR data **136**, if the partners **508** have consent to the scopes **506A-C**. For example, the FHIR data **136A** can include scopes A-F. The scope **506A-F** can include consent for a first partner **508A** and a second partner **508**, that correspond to a first partner server **114A** and second partner server **114B**, respectively. The profiles **132** can store the list of partners **508** for which the user **202** corresponding to the user ID **502** may transmit the FHIR data **136** from the healthcare **504** to the partner **508** (or partner sever **114**). For example, the user **202** with user ID **502A** can have a first partner **508B** and a second partner **508D** located within the FHIR data **136A** within the healthcare **504D**.

[0054] In some embodiments, the data structure **500** can include consent for each partner **508**. The user **202** determines, assigns, or establishes the consent for each partner **508** when the user **202** creates a profile **132** or during the duration of using the application **106**. The consent can restrict or allow the transmission of the FHIR data to the partner **508**. For example, FHIR data **136B** in healthcare **504C** can include a first partner **508B** and a second partner **508E**. The user **202** may give consent to the first partner **508B**. In this manner, the data management system **102** may transmit the FHIR data **136B** to first partner **508** unless consent is given to the second partner **508**. Each user **202** can establish any number of consent for the partners **508**. For example, a first user ID **502** may give consent to two partners **508A-B**, whereas a second user ID **502B** may give consent to three partners **508B-D**.

[0055] The consent may identify or specify a type of FHIR data **136** or a portion of FHIR data **136** to transmit to the partner server **114** based on the scopes **506**. For example, the consent may specify that the partner **504B** access scopes **506G-I** which correspond to allergies to medications, demographics, and blood pression of the user ID **502B**, whereas the partner **504K** may not have access to any of the scopes **606**. Using the consent, the user's **202** can restrict and manage FHIR data **136** accessible by the partners **508**. The FHIR manager **122** can provide a user interface to provide access control and an option to revoke consent to the one or more partners **508** as shown in FIGS. **6A-6B**.

[0056] FIG. **6A** depicts an example user interface **600** for managing access for the partners **508**. The user interface **600** can include one or more partners **508**. Each partner **508** can include a partner name, a creation date, and a link to the provider portal. The interaction manager **124** may monitor the user interface to detect an interaction **204** with the option to revoke consent as shown in user interface **630** of FIG. **6B**. In this manner, the interaction **204** may include a button press or keyboard interaction. In response to the interaction **204**, the FHIR manager **122** may update the one or more data structures **500** to remove the one or more partners **508** from the respective scope **506**. For example, the user **202** may revoke access from partner **508C**. The FHIR manager **122** may update the data structure **500** to remove the partner **508C** from the scopes **506** of the FHIR data **136**.

[0057] Referring now to FIG. **7**, FIG. **7** depicts an example **700** of an exchange of requests between the data management system **102** and the partner server **114**. The partner server **114** may transmit one or more requests **702A-N** (generally referred to as a “request **702**” or as “requests **702**” to the data management system **102**. The partner server **114** may use to request **702** to retrieve an access code **212** that the user **202** can exchange for an access token **134**. The access code **212** may provide the partner server **114** access to the FHIR data **136** associated with the data sources **112**. During this process, the data management system **102** can transmit scopes **704A-N** (generally referred to as

a “scope **704**” or as “scopes **704**”) to one or more partner servers **114**. The scopes **704** can identify, indicate, or determine one or more portions of the FHIR data **136** to transmit to the partner server **114**. For example, the FHIR data **136** can include the medical history associated with the user's **202** diagnosis of diabetes. The scope **704** may identify the medication history associated with the users **202** diabetes. The scopes **704** are the same as the scopes **506**. In another example, a second partner server **114B** can transmit the request **702B** to the data management system **102** at the same time as the partner server **114A**. The data management system **102** may follow a similar process as described above and transmit the scope **704B** to the partner server **114B**. Therefore, the data management system **102** can handle multiple interactions with multiple partner servers **114**.

[0058] In some embodiments, the partner server **114** can transmit a subsequent request **702B** to the data management system **102**. The subsequent request **702** can indicate more FHIR data **136** for the user **202**. The authorization manager **128** may verify that the partner server **114** has consent before authorizing the transmission of the scope **704**. Using the consent, the FHIR manager **122** may determine, identify, or access a second portion of data associated with the FHIR data **136** and transmit the data within the scope **704**.

[0059] The data management system **102** may exchange the access code **212** for a usable access token **134** corresponding to the respective data source **112**. For example, to access the user's **202** demographics for the data source **112G**, the partner server **114** may exchange the access code **212** for the access token **134** that grants access to the demographics of data source **112G**. Upon acceptance of the access token **134**, the data source manager **126** may transmit the demographics of the user **202** to the partner server **114**. The partner server **114** may store the demographics of the user **202** in the partner database **116** for reference in the future. The partner database **116** may store the access token **134** increase the speed of receiving user data from the data source **112**.

[0060] Prior to transmitting the FHIR data **136** to the partner server **114**, the authorization manager may verify, identify, or determine that the partner server **114** has consent to access the FHIR data **136**. During the verification process, the authorization manager **128** may retrieve the profile **132** corresponding to the user **202** and access the data structure **500** to parse each partner **508** to search for the consent. In this manner, the data management system **102** can improve the security of the FHIR data **136** by refusing to transmit any portions of data unless the partners **508** or partner server **114** obtain consent.

[0061] FIG. **8** depicts a flowchart method **800** for implementing secure access to the data. The method **800** can include at operation **805**, receiving, by a data management system **102**, a request to access data. The data management system **102** may receive the request responsive to the interaction manager **124** detecting an interaction **204** of a user **202** with an actionable object **302** within an information resource **108** of an application **106** on a user device **104**. The actionable object presented within the information resource **108** of the application **106** is presented via a software development kit (SDK) **110** of the data management system **102**. The method can include, receiving, by an authorization manager **128**, authentication credentials from the client device. The authentication credentials including a client identifier and a password corresponding to the client device. The method can include, at operation **810**, authenticating, by the authorization manager **128**, the user **202** of the user device **104**. The authorization manager **128** may authenticate the user **202** in response to the data management system **102** receiving the authentication credentials from the user device **104**.

[0062] At operation **815**, the method can include, providing, by a data source manager **126** to a data source **112**, a request to grant the data management system **102** access to user data stored at the one or more data sources. The request can include an identifier identifying the data management system **102**. The method can include, at operation **820**, receiving, by the data source manager **126**, an access token **134** to allow the data management system **102** to retrieve user data of the user **202** from the one or more data sources **112**. The method can include, at operation **825**, transmitting, by an application programming interface (API) manager **120**, via one or more of the one or more data

sources **112**, a request to access the user data **116** of the user **202** using the access token **134** assigned to the data management system **102**.

[0063] At operation **830**, the method can include, receiving, by a FHIR manager **122**, the user data **116** via the one or more APIs responsive to transmitting the request to access the user data **116**. At operation, **835**, the method can include, storing, by the FHIR manager **122** and the profile generator **130**, the user data **116** in one or more data structures **500** managed by the data management system **102**. The method can include, storing, by the FHIR manager **122**, consent data about the user data **116** in one or more data structures **500** managed by the data management system **102** based on a partner identifier. The consent data may indicate that the data management system **102** is configured to transmit data to the one or more servers **114** of the application **106**. The method can include, providing, by the authorization manager **128** or the interaction manager **124**, a user interface to the user **202** of the user device **104**. Responsive to the interaction **204** with the user interface to revoke consent to share the user data **116** of the user **202** with the application **106**, update the one or more data structures **500** managed by the data management system **102** to restrict the data management system from sharing the user data **116** with the application **106**.

[0064] At operation **840**, the method can include transmitting, by the data management system **102**, a portion (e.g., scopes **704**) of the user data **116** of the user **202** to the one or more servers (e.g., partner servers **114**) of the provider portal (e.g., partners **508**, application **106**) based on one or more values corresponding to a type of data obtained at registration of the application **106** with the data management system **102**. The method can include, transmitting, by the data management system **102**, responsive to a subsequent data request **702** from the one or more servers **114** of the application **106**, a second portion of the user data **116** of the user **202** to the application **106** based on the one or more values corresponding to the type of data obtained at registration of the application **106**.

[0065] Various operations described herein can be implemented on computer systems, which can be of generally conventional design. FIG. **9** shows a simplified block diagram of a representative server system **900** and client computer system **914** usable to implement certain embodiments of the technical solutions described herein. In various embodiments, server system **900** or similar systems can implement services or servers described herein or portions thereof. Client computer system **914** or similar systems can implement clients described herein. Each of the systems **100**, **200** and others described herein can be similar to the server system **900**.

[0066] Server system **900** can have a modular design that incorporates a number of modules **902** (e.g., blades in a blade server embodiment); while two modules **902** are shown, any number can be provided. Each module **902** can include processing unit(s) **904** and local storage **906**.

[0067] Processing unit(s) **904** can include a single processor, which can have one or more cores, or multiple processors. In some embodiments, processing unit(s) **904** can include a general-purpose primary processor as well as one or more special-purpose co-processors such as graphics processors, digital signal processors, or the like. In some embodiments, some or all processing units **904** can be implemented using customized circuits, such as application specific integrated circuits (ASICs) or field programmable gate arrays (FPGAs). In some embodiments, such integrated circuits execute instructions that are stored on the circuit itself. In other embodiments, processing unit(s) **904** can execute instructions stored in local storage **906**. Any type of processors in any combination can be included in processing unit(s) **904**.

[0068] Local storage **906** can include volatile storage media (e.g., conventional DRAM, SRAM, SDRAM, or the like) and/or non-volatile storage media (e.g., magnetic or optical disk, flash memory, or the like). Storage media incorporated in local storage **906** can be fixed, removable or upgradeable as desired. Local storage **906** can be physically or logically divided into various subunits such as a system memory, a read-only memory (ROM), and a permanent storage device. The system memory can be a read-and-write memory device or a volatile read-and-write memory, such as dynamic random-access memory. The system memory can store some or all of the

instructions and data that processing unit(s) **904** need at runtime. The ROM can store static data and instructions that are needed by processing unit(s) **904**. The permanent storage device can be a non-volatile read-and-write memory device that can store instructions and data even when module **902** is powered down. The term “storage medium” as used herein includes any medium in which data can be stored indefinitely (subject to overwriting, electrical disturbance, power loss, or the like) and does not include carrier waves and transitory electronic signals propagating wirelessly or over wired connections.

[0069] In some embodiments, local storage **906** can store one or more software programs to be executed by processing unit(s) **904**, such as an operating system and/or programs implementing various server functions such as functions of the system **100** of FIG. **1** or any other system described herein.

[0070] “Software” refers generally to sequences of instructions that, when executed by processing unit(s) **904** cause server system **900** (or portions thereof) to perform various operations, thus defining one or more specific machine embodiments that execute and perform the operations of the software programs. The instructions can be stored as firmware residing in read-only memory and/or program code stored in non-volatile storage media that can be read into volatile working memory for execution by processing unit(s) **904**. Software can be implemented as a single program or a collection of separate programs or program modules that interact as desired. From local storage **906** (or non-local storage described below), processing unit(s) **904** can retrieve program instructions to execute and data to process in order to execute various operations described above.

[0071] In some server systems **900**, multiple modules **902** can be interconnected via a bus or other interconnect **908**, forming a local area network that supports communication between modules **902** and other components of server system **900**. Interconnect **908** can be implemented using various technologies including server racks, hubs, routers, etc.

[0072] A wide area network (WAN) interface **910** can provide data communication capability between the local area network (interconnect **908**) and a larger network, such as the Internet. Conventional or other activities technologies can be used, including wired (e.g., Ethernet, IEEE 902.3 standards) and/or wireless technologies (e.g., Wi-Fi, IEEE 902.11 standards).

[0073] In some embodiments, local storage **906** is intended to provide working memory for processing unit(s) **904**, providing fast access to programs and/or data to be processed while reducing traffic on interconnect **908**. Storage for larger quantities of data can be provided on the local area network by one or more mass storage subsystems **912** that can be connected to interconnect **908**. Mass storage subsystem **912** can be based on magnetic, optical, semiconductor, or other data storage media. Direct attached storage, storage area networks, network-attached storage, and the like can be used. Any data stores or other collections of data described herein as being produced, consumed, or maintained by a service or server can be stored in mass storage subsystem **912**. In some embodiments, additional data storage resources may be accessible via WAN interface **910** (potentially with increased latency).

[0074] Server system **900** can operate in response to requests received via WAN interface **910**. For example, one of modules **902** can implement a supervisory function and assign discrete tasks to other modules **902** in response to received requests. Conventional work allocation techniques can be used. As requests are processed, results can be returned to the requester via WAN interface **910**. Such operation can generally be automated. Further, in some embodiments, WAN interface **910** can connect multiple server systems **900** to each other, providing scalable systems capable of managing high volumes of activity. Conventional or other techniques for managing server systems and server farms (collections of server systems that cooperate) can be used, including dynamic resource allocation and reallocation.

[0075] Server system **900** can interact with various user-owned or user-operated devices via a wide-area network such as the Internet. An example of a user-operated device is shown in FIG. **8** as client computing system **914**. Client computing system **914** can be implemented, for example, as a

consumer device such as a smartphone, other mobile phone, tablet computer, wearable computing device (e.g., smart watch, eyeglasses), desktop computer, laptop computer, and so on.

[0076] For example, client computing system **914** can communicate via WAN interface **910**. Client computing system **914** can include conventional computer components such as processing unit(s) **916**, storage device **918**, network interface **920**, user input device **922**, and user output device **924**. Client computing system **914** can be a computing device implemented in a variety of form factors, such as a desktop computer, laptop computer, tablet computer, smartphone, other mobile computing device, wearable computing device, or the like.

[0077] Processor **916** and storage device **918** can be similar to processing unit(s) **904** and local storage **906** described above. Suitable devices can be selected based on the demands to be placed on client computing system **914**; for example, client computing system **914** can be implemented as a “thin” client with limited processing capability or as a high-powered computing device. Client computing system **914** can be provisioned with program code executable by processing unit(s) **916** to enable various interactions with server system **900** of a message management service such as accessing messages, performing actions on messages, and other interactions described above. Some client computing systems **914** can also interact with a messaging service independently of the message management service.

[0078] Network interface **920** can provide a connection to a wide area network (e.g., the Internet) to which WAN interface **910** of server system **900** is also connected. In various embodiments, network interface **920** can include a wired interface (e.g., Ethernet) and/or a wireless interface implementing various RF data communication standards such as Wi-Fi, Bluetooth, or cellular data network standards (e.g., 3G, 4G, LTE, etc.).

[0079] User input device **922** (or user devices **104**) can include any device (or devices) via which a user can provide signals to client computing system **914**; client computing system **914** can interpret the signals as indicative of particular user requests or information. In various embodiments, user input device **922** can include any or all of a keyboard, touch pad, touch screen, mouse or other pointing device, scroll wheel, click wheel, dial, button, switch, keypad, microphone, and so on.

[0080] User output device **924** (or user devices **104**) can include any device via which client computing system **914** can provide information to a user. For example, user output device **924** can include a display to display images generated by or delivered to client computing system **914**. The display can incorporate various image generation technologies, e.g., a liquid crystal display (LCD), light-emitting diode (LED) including organic light-emitting diodes (OLED), projection system, cathode ray tube (CRT), or the like, together with supporting electronics (e.g., digital-to-analog or analog-to-digital converters, signal processors, or the like). Some embodiments can include a device such as a touchscreen that function as both input and output device. In some embodiments, other user output devices **924** can be provided in addition to or instead of a display. Examples include indicator lights, speakers, tactile “display” devices, printers, and so on.

[0081] Some embodiments include electronic components, such as microprocessors, storage and memory that store computer program instructions in a computer readable storage medium. Many of the features described in this specification can be implemented as processes that are specified as a set of program instructions encoded on a computer readable storage medium. When these program instructions are executed by one or more processing units, they cause the processing unit(s) to perform various operation indicated in the program instructions. Examples of program instructions or computer code include machine code, such as is produced by a compiler, and files including higher-level code that are executed by a computer, an electronic component, or a microprocessor using an interpreter. Through suitable programming, processing unit(s) **904** and **916** can provide various functionality for server system **900** and client computing system **914**, including any of the functionality described herein as being performed by a server or client, or other functionality associated with message management services.

[0082] It will be appreciated that server system **900** and client computing system **914** are

illustrative and that variations and modifications are possible. Computer systems used in connection with embodiments of the technical solutions described herein can have other capabilities not specifically described here. Further, while server system **900** and client computing system **914** are described with reference to particular blocks, it is to be understood that these blocks are defined for convenience of description and are not intended to imply a particular physical arrangement of component parts. For instance, different blocks can be but need not be located in the same facility, in the same server rack, or on the same motherboard. Further, the blocks need not correspond to physically distinct components. Blocks can be configured to perform various operations, e.g., by programming a processor or providing appropriate control circuitry, and various blocks might or might not be reconfigurable depending on how the initial configuration is obtained. Embodiments of the technical solutions described herein can be realized in a variety of apparatus including electronic devices implemented using any combination of circuitry and software.

[0083] While the disclosure has been described with respect to specific embodiments, one skilled in the art will recognize that numerous modifications are possible. For instance, although specific examples of rules (including triggering conditions and/or resulting actions) and processes for generating suggested rules are described, other rules and processes can be implemented.

Embodiments of the disclosure can be realized using a variety of computer systems and communication technologies including but not limited to specific examples described herein.

[0084] Embodiments of the technical solutions described herein can be realized using any combination of dedicated components and/or programmable processors and/or other programmable devices. The various processes described herein can be implemented on the same processor or different processors in any combination. Where components are described as being configured to perform certain operations, such configuration can be accomplished, e.g., by designing electronic circuits to perform the operation, by programming programmable electronic circuits (such as microprocessors) to perform the operation, or any combination thereof. Further, while the embodiments described above may make reference to specific hardware and software components, those skilled in the art will appreciate that different combinations of hardware and/or software components may also be used and that particular operations described as being implemented in hardware might also be implemented in software or vice versa.

[0085] Computer programs incorporating various features of the technical solutions described herein may be encoded and stored on various computer readable storage media; suitable media include magnetic disk or tape, optical storage media such as compact disk (CD) or DVD (digital versatile disk), flash memory, and other non-transitory media. Computer readable media encoded with the program code may be packaged with a compatible electronic device, or the program code may be provided separately from electronic devices (e.g., via Internet download or as a separately packaged computer-readable storage medium).

[0086] Thus, although the disclosure has been described with respect to specific embodiments, it will be appreciated that the disclosure is intended to cover all modifications and equivalents within the scope of the following claims.

Claims

1. A system comprising: one or more processors of a data management system coupled to memory, the one or more processors of the data management system configured to: receive, responsive to an interaction with an actionable object presented within an information resource of a provider portal at a client device of a user, a request to access data; authenticate, responsive to receiving authentication credentials from the client device, the user of the client device; provide, to a data source, a request to grant, to the data management system, access to user data of the user stored at one or more data sources, the request including a client identifier identifying the data management system; receive, responsive to the request to grant, an access token to allow the data management

system to retrieve user data of the user from the one or more data sources; transmit, via one or more application programming interfaces (API) of the one or more data sources, a request to access the user data of the user using the access token assigned to the data management system; receive the user data via the one or more APIs responsive to transmitting the request to access the user data; store the user data in one or more data structures managed by the data management system; transmit, responsive to a request from one or more servers of the provider portal, a portion of the user data of the user to the one or more servers of the provider portal based on one or more values corresponding to a type of data obtained at registration of the provider portal with the data management system; receive, responsive to a second interaction with a second actionable object presented within a second information resource of a second provider portal at the client device, a data access request to access data; authenticate, responsive to receiving the authentication credentials from the client device, the user of the client device; provide, to a second data source of the one or more data sources, a second request to grant, to the data management system, access to user data of the user stored at the second data source, the second request including the client identifier identifying the data management system; receive, responsive to the second request to grant, the second access token to allow the data management system to retrieve the user data of the user from the one or more data sources; transmit, via the one or more application programming interfaces (API) of the one or more data sources, a second request to access the user data of the user using the second access token assigned to the data management system; receive the user data via the one or more APIs responsive to transmitting the second request to access the user data of the user; store the user data of the user in the one or more data structures managed by the data management system; and transmit, responsive to a second request from one or more servers of the second provider portal, a second portion of the user data of the user to the one or more servers of the second provider portal based on one or more values corresponding to the type of data obtained at registration of the second provider portal with the data management system.

2. The system of claim 1, wherein the user is a first user and the client device is a first client device and wherein the one or more processors are further configured to: receive, responsive to a second interaction with the actionable object presented within the information resource of the provider portal at a second client device of a second user, a data access request to access data; authenticate, responsive to receiving second authentication credentials from the second client device, the second user of the second client device; provide, to a second data source, a second request to grant, to the data management system, access to second user data of the second user stored at the one or more data sources, the second request including the client identifier identifying the data management system; receive, responsive to the second request to grant, a second access token to allow the data management system to retrieve second user data of the second user from the one or more data sources; transmit, via the one or more application programming interfaces (API) of the one or more data sources, a second request to access the second user data of the second user using the second access token assigned to the data management system; receive the second user data via the one or more APIs responsive to transmitting the second request to access the second user data; store the second user data in one or more second data structures managed by the data management system; and transmit, responsive to a second request from the one or more servers of the provider portal, a respective portion of the second user data of the second user to the one or more servers of the provider portal based on the one or more values corresponding to the type of data obtained at registration of the first provider portal with the data management system.

3. The system of claim 1, wherein the type of data includes a medical history, medical records, medical conditions, immunization records, clinical notes, or medications.

4. The system of claim 1, wherein the one or more processors are further configured to: receive, responsive to a second interaction with a second actionable object presented within a second information resource of a second provider portal at a second client device, a data access request to access data; authenticate, responsive to receiving second authentication credentials from the second

client device, a second user of the second client device; provide, to a second data source of the one or more data sources, a second request to grant, to the data management system, access to second user data of the second user stored at the one or more data sources, the second request to grant including the client identifier identifying the data management system; receive, responsive to the second request to grant, a second access token to allow the data management system to retrieve second user data of the second user from the one or more second data sources; transmit, via the one or more application programming interfaces (API) of the one or more data sources, a second request to access the second user data of the second user using the access token assigned to the data management system; receive the second user data via the one or more APIs responsive to transmitting the second request to access the second user data; store the second user data of the second user in one or more second data structures managed by the data management system; and transmit, responsive to a second request from one or more servers of the second provider portal, portions of the second user data of the second user to the one or more servers of the second provider portal based on one or more values corresponding to a type of data obtained at registration of the second provider portal with the data management system.

5. The system of claim 1, wherein the one or more processors are further configured to transmit, responsive to a subsequent data request from the one or more servers of the provider portal, a second portion of the user data of the user to the provider portal based on the one or more values corresponding to the type of data obtained at registration of the provider portal.

6. The system of claim 1, wherein the one or more processors are further configured to receive the authentication credentials from the client device, the authentication credentials including a client identifier and a password corresponding to the client device.

7. The system of claim 1, wherein the one or more processors are further configured to transmit, responsive to the request from the one or more servers of the provider portal, the portion of the user data of the user to the client device based on one or more values corresponding to the type of data obtained at registration of the provider portal.

8. The system of claim 1, wherein the one or more processors are further configured to store consent data about the user data in one or more data structures managed by the data management system based on a partner identifier, the consent data indicating that the data management system is configured to transmit data to the one or more servers of the provider portal.

9. The system of claim 8, wherein the one or more processors are further configured to provide a user interface to the user of the client device, and responsive to an interaction with the user interface to revoke consent to share the user data of the user with the provider portal, update the one or more data structures managed by the data management system to restrict the data management system from sharing the user data with the provider portal.

10. The system of claim 1, wherein the actionable object presented within the information resource of the provider portal is presented via a software development kit (SDK) of the data management system.

11. A computer implemented method comprising: receiving, by one or more processors, responsive to an interaction with an actionable object presented within an information resource of a provider portal at a client device of a user, a request to access data; authenticating, by the one or more processors, responsive to receiving authentication credentials from the client device, the user of the client device; providing, by the one or more processors, to a data source, a request to grant, to a data management system, access to user data of the user stored at one or more data sources, the request including a client identifier identifying the data management system; receiving, by the one or more processors, responsive to the request to grant, an access token to allow the data management system to retrieve user data of the user from the one or more data sources; transmitting, by the one or more processors, via one or more application programming interfaces (API) of the one or more data sources, a request to access the user data of the user using the access token assigned to the data management system; receiving, by the one or more processors, the user

data via the one or more APIs responsive to transmitting the request to access the user data; storing, by the one or more processors, the user data in one or more data structures managed by the data management system; transmitting, by the one or more processors, responsive to a request from one or more servers of the provider portal, a portion of the user data of the user to the one or more servers of the provider portal based on one or more values corresponding to a type of data obtained at registration of the provider portal with the data management system; receiving, by the one or more processors, responsive to a second interaction with a second actionable object presented within a second information resource of a second provider portal at the client device, a data access request to access data; authenticating, by the one or more processors, responsive to receiving the authentication credentials from the client device, a first user of the client device; providing, by the one or more processors, to a second data source of the one or more data sources, a second request to grant, to the data management system, access to user data of the first user stored at the second data source, the second request including the client identifier identifying the data management system; receiving, by the one or more processors, responsive to the second request to grant, a second access token to allow the data management system to retrieve the user data of the first user from the one or more data sources; transmitting, by the one or more processors, via the one or more application programming interfaces (API) of the one or more data sources, a second request to access the user data of the first user using the second access token assigned to the data management system; receiving, by the one or more processors, the user data via the one or more APIs responsive to transmitting the second request to access the user data of the first user; storing, by the one or more processors, the user data of the first user in the one or more data structures managed by the data management system; and transmitting, by the one or more processors, responsive to a second request from one or more servers of the second provider portal, a second portion of the user data of the first user to the one or more servers of the second provider portal based on one or more values corresponding to the type of data obtained at registration of the second provider portal with the data management system.

12. The method of claim 11, wherein the user is a first user and the client device is a first client device, further comprising: receiving, by the one or more processors, responsive to a second interaction with the actionable object presented within the information resource of the provider portal at a second client device of a second user, a data access request to access data; authenticating, by the one or more processors, responsive to receiving second authentication credentials from the second client device, the second user of the second client device; providing, by the one or more processors, to a second data source, a second request to grant, to the data management system, access to second user data of the second user stored at the one or more data sources, the second request including the client identifier identifying the data management system; receiving, by the one or more processors, responsive to the second request to grant, a second access token to allow the data management system to retrieve second user data of the second user from the one or more data sources; transmitting, by the one or more processors, via the one or more application programming interfaces (API) of the one or more data sources, a second request to access the second user data of the second user using the second access token assigned to the data management system; receiving, by the one or more processors, the second user data via the one or more APIs responsive to transmitting the second request to access the second user data; storing, by the one or more processors, the second user data in one or more second data structures managed by the data management system; and transmitting, by the one or more processors, responsive to a second request from the one or more servers of the provider portal, a respective portion of the second user data of the second user to the one or more servers of the provider portal based on the one or more values corresponding to the type of data obtained at registration of the first provider portal with the data management system.

13. The method of claim 11, wherein the type of data includes a medical history, medical records, medical conditions, immunization records, clinical notes, or medications.

14. The method of claim 11, further comprising: receiving, by the one or more processors, responsive to a second interaction with a second actionable object presented within a second information resource of a second provider portal at a second client device, a data access request to access data; authenticating, by the one or more processors, responsive to receiving second authentication credentials from the second client device, a second user of the second client device; providing, by the one or more processors, to a second data source of the one or more data sources, a second request to grant, to the data management system, access to second user data of the second user stored at the one or more data sources, the second request to grant including the client identifier identifying the data management system; receiving, by the one or more processors, responsive to the second request to grant, a second access token to allow the data management system to retrieve second user data of the second user from the one or more second data sources; transmitting, by the one or more processors, via the one or more application programming interfaces (API) of the one or more data sources, a second request to access the second user data of the second user using the access token assigned to the data management system; receiving, by the one or more processors, the second user data via the one or more APIs responsive to transmitting the second request to access the second user data; storing, by the one or more processors, the second user data of the second user in one or more second data structures managed by the data management system; and transmitting, by the one or more processors, responsive to a second request from one or more servers of the second provider portal, portions of the second user data of the second user to the one or more servers of the second provider portal based on one or more values corresponding to a type of data obtained at registration of the second provider portal with the data management system.

15. The method of claim 11, further comprising: transmitting, by the one or more processors, responsive to a subsequent data request from the one or more servers of the provider portal, a second portion of the user data of the user to the provider portal based on the one or more values corresponding to the type of data obtained at registration of the provider portal.

16. The method of claim 11, further comprising: receiving, by the one or more processors, the authentication credentials from the client device, the authentication credentials including a client identifier and a password corresponding to the client device.

17. The method of claim 11, further comprising: transmitting, by the one or more processors, responsive to the request from the one or more servers of the provider portal, the portion of the user data of the user to the client device based on one or more values corresponding to the type of data obtained at registration of the provider portal.

18. The method of claim 11, further comprising: storing, by the one or more processors, consent data about the user data in one or more data structures managed by the data management system based on a partner identifier, the consent data indicating that the data management system is configured to transmit data to the one or more servers of the provider portal.

19. The method of claim 18, further comprising: providing, by the one or more processors, a user interface to the user of the client device, and responsive to an interaction with the user interface to revoke consent to share the user data of the user with the provider portal, update the one or more data structures managed by the data management system to restrict the data management system from sharing the user data with the provider portal.

20. The method of claim 11, wherein the actionable object presented within the information resource of the provider portal is presented via a software development kit (SDK) of the data management system.
